

$$V_{SP} = \frac{\sqrt{\frac{\beta_n}{4\beta_p}} \cdot V_{THN} + (VDD - V_{THP})}{1 + \sqrt{\frac{\beta_n}{4\beta_p}}} \quad (12.6)$$

or in general for an n-input NAND gate (see Fig. 12.2), we get

$$V_{SP} = \frac{\sqrt{\frac{\beta_n}{N^2 \cdot \beta_p}} \cdot V_{THN} + (VDD - V_{THP})}{1 + \sqrt{\frac{\beta_n}{N^2 \cdot \beta_p}}} \quad (12.7)$$

These equations are derived under the assumption that all inputs are tied together. If, for example, only one input is switching, the V_{SP} will vary from what is calculated using Eq. (12.7) (assuming the single input switching does indeed cause the gate's output to switch). This equation is used to show why NAND gates are preferred in CMOS design. If equal-sized NMOS and PMOS devices are used, then, since the mobility of the hole is less than the mobility of the electron, $\beta_n > \beta_p$. Using NMOS devices in series and PMOS in parallel (as in the NAND gate) makes it easier to design a logic gate with the ideal switching point voltage of $VDD/2$.

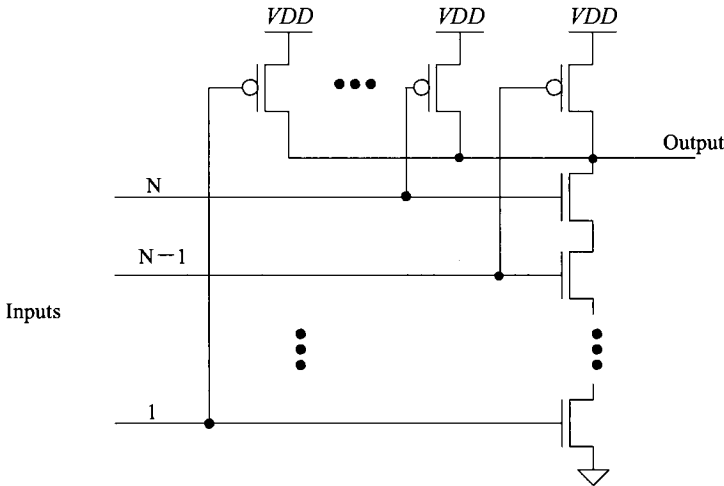


Figure 12.2 Schematic of an n-input NAND gate.

Example 12.1

Determine V_{SP} by hand calculations and compare to a SPICE simulation for a three-input NAND gate using 10/1 devices in the long-channel CMOS process used in this book, see Table 6.2. Compare the hand calculations to simulation results.

The switching point voltage is determined by calculating the transconductance ratio of the gate, or

$$\sqrt{\frac{\beta_n}{N^2 \beta_p}} = \sqrt{\frac{\frac{120 \mu A/V^2 \cdot 10}{1}}{9 \cdot \frac{40 \mu A/V^2 \cdot 10}{1}}} = 0.58$$

and then using Eq. (12.7),

$$V_{SP} = \frac{0.58 \cdot (0.8) + (5 - 0.9)}{1 + 0.58} = 2.9 \text{ V}$$

The SPICE simulation results are shown in Fig. 12.3. The simulation also gives a V_{SP} of approximately 2.9 V. ■

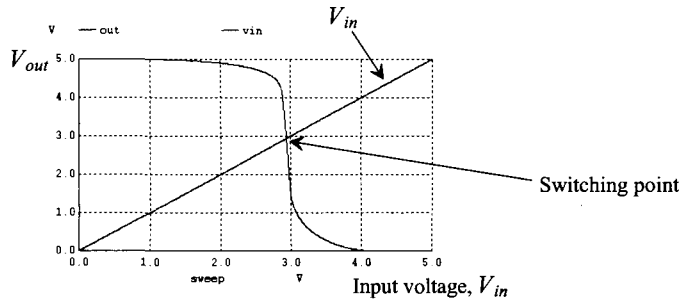


Figure 12.3 VTCs of the three-input minimum-size (using 10/1 MOSFETs) NAND gate.

12.1.2 DC Characteristics of the NOR gate

Following a similar analysis for the n-input NOR gate (see Fig. 12.4) gives a switching point voltage of

$$V_{SP} = \frac{\sqrt{\frac{N^2 \cdot \beta_n}{\beta_p}} \cdot V_{THN} + (V_{DD} - V_{THP})}{1 + \sqrt{\frac{N^2 \cdot \beta_n}{\beta_p}}} \quad (12.8)$$

Example 12.2

Compare the switching point voltage of a three-input NOR gate made from minimum-size MOSFETs to that of the three-input NAND gate of Ex. 12.1. Comment on which gate's V_{SP} is closer to ideal, that is, $V_{SP} = V_{DD}/2$.

The transconductance ratio is calculated as

$$\sqrt{N^2 \cdot \frac{\beta_n}{\beta_p}} = \sqrt{9 \cdot \frac{\frac{120 \mu A/V^2 \cdot 10}{1}}{\frac{40 \mu A/V^2 \cdot 10}{1}}} = 5.2$$

The V_{SP} of the minimum-size three-input NOR gate is 1.33 V, while the V_{SP} of the minimum-size three-input NAND gate was calculated to be 2.9 V. For an ideal

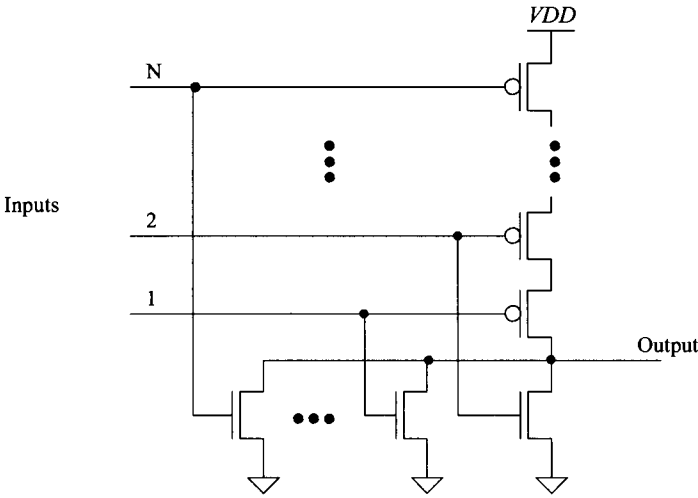


Figure 12.4 Schematic of an n-input NOR gate.

gate, $V_{SP} = 2.5$ V, so that the NAND gate is closer to ideal than the NOR gate. In CMOS digital design, the NAND gate is used most often. This is due to the DC characteristics, better noise margins, and the dynamic characteristics. We will also see shortly that the NAND gate has better transient characteristics than the NOR gate. ■

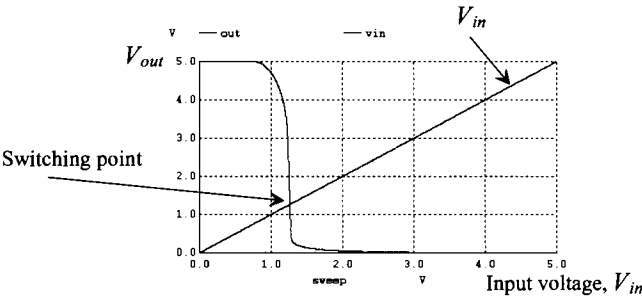


Figure 12.5 VTCs of the three-input minimum-size (using 10/1 MOSFETs) NOR gate.

A Practical Note Concerning V_{SP} and Pass Gates

Reviewing Fig. 10.19, we see that passing a logic signal through a pass gate (PG) can result in a reduction in the logic signal’s amplitude. Using an NMOS PG, for example, results in an output signal swing from ground to $V_{DD} - V_{THN}$. If this logic signal is connected to an inverter or a logic gate, we will want to set the V_{SP} to maximize the noise margins. Using the NMOS PG, our inverter/gate would have a $V_{SP} = (V_{DD} - V_{THN})/2$.

12.2 Layout of the NAND and NOR Gates

Layout of the three-input minimum-size NOR and NAND gates is shown in Fig. 12.6, using the standard-cell frame. MOSFETs in series, for example, the NMOS devices in the NAND gate, are laid out using a single-drain area and a single-source implant area. The active area between the gate poly is shared between two devices. This has the effect of reducing the parasitic drain/source implant capacitances. MOSFETs in parallel, for example, the PMOS devices in the NOR gate, can share a drain area or a source area. The inputs of the gates are shown on the poly layer while the outputs of the two logic gates are on metal2. To make the inputs easy to connect to we would route them up to metal2 like the outputs. Note how metal1 is used inside the cell and horizontally to connect power and ground to the cells, while metal2 is used for vertical running wires (inputs and outputs).

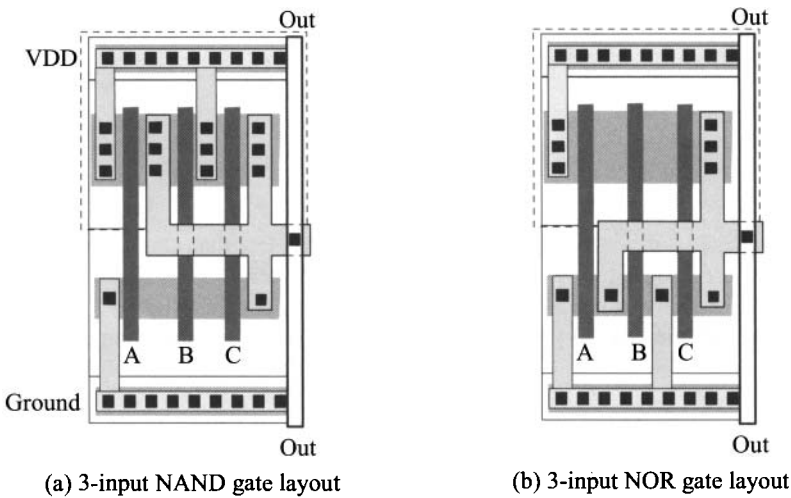


Figure 12.6 Layouts of NAND (a) and NOR (b) gates.

12.3 Switching Characteristics

In this section we discuss the switching characteristics of static logic gates.

Parallel Connection of MOSFETs

Consider the parallel connection of identical MOSFETs shown in Fig. 12.7 *with their gates tied together*. From the equivalent digital models, also shown, we can determine the propagation delay associated with this parallel connection of N MOSFETs as

$$t_{PLH} = 0.7 \cdot \frac{R_p}{N} \cdot (N \cdot C_{oxp}) = 0.7 \cdot R_p C_{oxp} \quad (12.9)$$

where $C_{oxp} = C'_{ox} \cdot W \cdot L \cdot (scale)^2$. With an external load capacitance, the low-to-high delay-time becomes

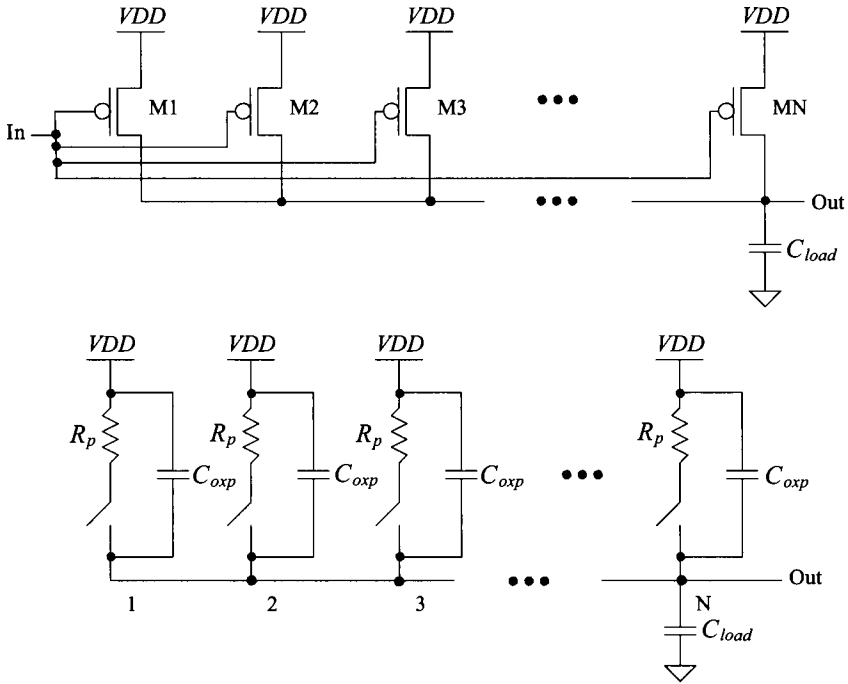


Figure 12.7 Parallel connection of MOSFETs and equivalent digital model.

$$t_{PLH} = 0.7 \cdot \frac{R_p}{N} \cdot (N \cdot C_{exp} + C_{load}) \quad (12.10)$$

This again assumes that the MOSFET's gates are tied together (all are switching at the same time). For NMOS devices in parallel, a similar analysis yields

$$t_{PHL} = 0.7 \cdot \frac{R_n}{N} \cdot (N \cdot C_{oxn} + C_{load}) \quad (12.11)$$

The load capacitance, C_{load} , consists of all capacitances on the output node except the output capacitances of the MOSFETs in parallel.

Series Connection of MOSFETs

Consider the series connection of identical NMOS devices shown in Fig. 12.8. We can *estimate* the intrinsic switching time of series-connected MOSFETs by

$$t_{PHL} = 0.35 \cdot R_n C_{oxn} \cdot N^2 \quad (12.12)$$

as discussed back in Sec. 10.2.2. With an external load capacitance, the high-to-low delay-time becomes

$$t_{PHL} = 0.35 \cdot R_n C_{oxn} \cdot N^2 + 0.7 \cdot N \cdot R_n \cdot C_{load} \quad (12.13)$$

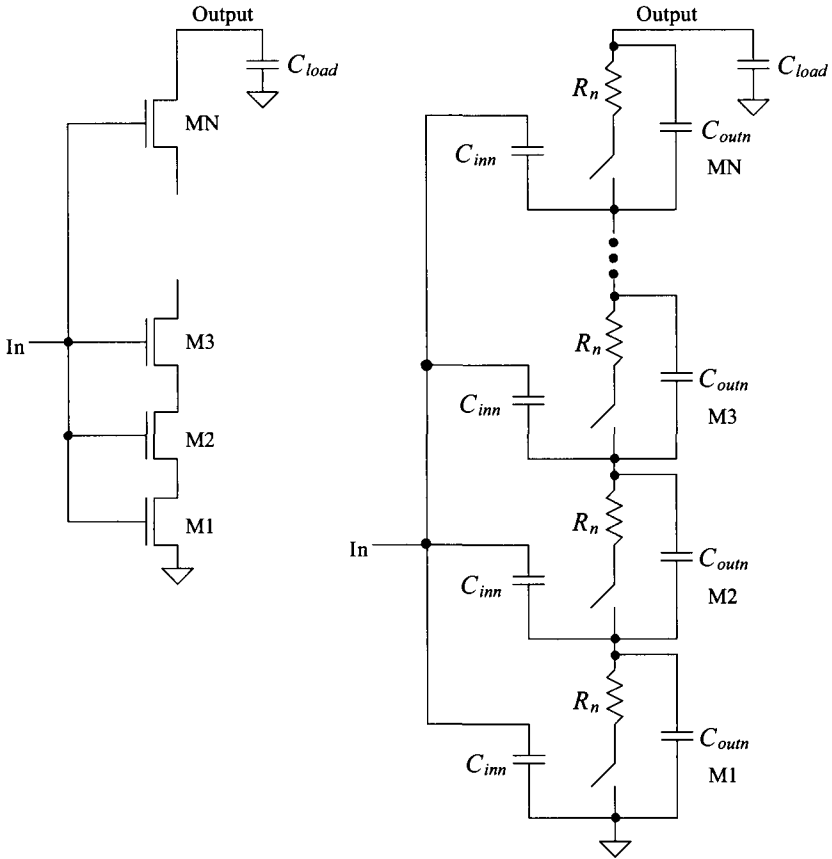


Figure 12.8 Series connection of MOSFETs and equivalent digital model.

For PMOS devices in series, a similar analysis yields

$$t_{PLH} = 0.35 \cdot R_p C_{oxp} \cdot N^2 + 0.7 \cdot N \cdot R_p \cdot C_{load} \quad (12.14)$$

These equations are approximations for the propagation delays which give results usually to within a factor of two of the measurements.

12.3.1 NAND Gate

Consider the n -input NAND gate of Fig. 12.9 driving a capacitive load C_{load} . The low-to-high propagation time, using Eq. (12.10), is

$$t_{PLH} = 0.7 \cdot \frac{R_p}{N} \left(N \cdot C_{outp} + \frac{C_{outn}}{N} + C_{load} \right) \quad (12.15)$$

where here C_{load} represents the capacitance external to the gate, whereas in Eq. (12.10) C_{load} represented the capacitance external to the parallel PMOS devices. If the load

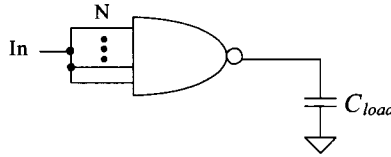


Figure 12.9 An n-input NAND gate driving a load capacitance.

capacitance is much greater than the output capacitance of the NAND gate, the low-to-high propagation time can be estimated by

$$t_{PLH} \approx 0.7 \cdot \frac{R_p}{N} \cdot C_{load} \quad (12.16)$$

The high-to-low propagation time, using Eq. (12.13), is given by

$$t_{PHL} = 0.7 \cdot N \cdot R_n \left[N \cdot C_{outn} + \frac{C_{outn}}{N} + C_{load} \right] + 0.35 \cdot R_n C_{oxn} \cdot N^2 \quad (12.17)$$

If C_{load} is much larger than the output capacitance of the NAND gate, then

$$t_{PHL} \approx 0.7 \cdot N \cdot R_n \cdot C_{load} \quad (12.18)$$

Example 12.3

Estimate the intrinsic propagation delays, $t_{PHL} + t_{PLH}$, of a three-input NAND gate made using 10/1 NMOS and 20/1 PMOS in the short-channel process. Estimate and simulate the delay when the gate is driving a load capacitance of 50 fF. Assume that the inputs are tied together.

Using the data in Table 10.2 and Eqs. (12.15) and (12.17),

$$t_{PLH} = 0.7 \cdot \frac{3.4k}{3} \cdot \left(3 \cdot 1.25 \text{ fF} + \frac{0.625 \text{ fF}}{3} + 50 \text{ fF} \right) = 43 \text{ ps}$$

and

$$t_{PHL} = 0.7 \cdot 3 \cdot 3.4k \cdot \left[3 \cdot 1.25 \text{ fF} + \frac{0.625 \text{ fF}}{3} + 50 \text{ fF} \right] + 0.35 \cdot 3.4k \cdot 0.625 \text{ fF} \cdot 9 = 393 \text{ ps}$$

The simulation results are seen in Fig. 12.10. ■

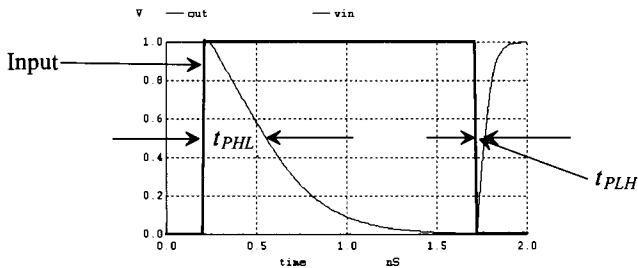


Figure 12.10 Simulating the operation of a 3-input NAND gate in 50 nm CMOS driving a 50 fF load capacitance.